SS2 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 5: NUCLEUS

1. DEFINITION OF NUCLEUS

The **nucleus** is the small, dense center of an atom where most of the atom's mass is concentrated. It was discovered through the gold foil experiment conducted by **Ernest Rutherford in 1911**. The nucleus is positively charged and occupies only a tiny fraction of the volume of the atom but contains nearly all its mass

2. CONSTITUENTS OF THE NUCLEUS

The nucleus is made up of two fundamental subatomic particles:

Particle	Symbol	Charge	Mass (approx.)	Description
Proton	p	$+1.6 \times 10^{-19}$ C	1 u	Positively charged; number of protons determines the element (atomic number Z)
Neutron	n	0	1 u	Electrically neutral; contributes to mass and nuclear stability

Together, protons and neutrons are called NUCLEONS.

3. NUCLEON NUMBER, PROTON NUMBER, ISOTOPES, AND NUCLIDES

3.1 Nucleon Number (Mass Number, A)

Total number of protons and neutrons in the nucleus.

Formula: $A = Z + N$

A = Nucleon number

Z = Number of protons (proton number)

N = Number of neutrons

3.2 Proton Number (Z)

Number of protons in the nucleus.

Defines the chemical identity of an atom (e.g., hydrogen has $Z = 1$).

3.3 Isotopes

Atoms of the **same element** (same proton number) but **different nucleon numbers**.

Example: Hydrogen isotopes:

Protium (^1H)

Deuterium (^2H)

Tritium (^3H)

3.4 Nuclides

A nuclide is a specific isotope of an element characterized by its proton and neutron numbers.

Represented as: ᴬ_Z X

Example: $^{14}_{6}C$ represents a carbon nuclide with 6 protons and 8 neutrons.

4. NUCLEAR PROPERTIES

4.1 Mass Defect (Δm)

The difference between the sum of the masses of individual nucleons and the actual mass of the nucleus.

Indicates that some mass has been converted into energy that binds the nucleons together.

4.2 Binding Energy (BE)

The energy required to split a nucleus into its individual protons and neutrons.

Reflects the stability of the nucleus

Calculated using **Einstein's mass-energy equivalence**:

$$E = \Delta m \cdot c$$

E = Binding energy (in joules or MeV)

Δm = Mass defect (in kg or u)

c = Speed of light (3×10^8 m/s)

4.3 Nuclear Stability

Stable nuclei have a balanced ratio of protons to neutrons.

Too many or too few neutrons can lead to **radioactive decay**.

4.4 Half-Life (T)

The half-life of a substance is defined as the time taken for half of the atoms in any given substance to decay.

Formula: $T = 0.692 / \lambda$

T = half-life

λ = decay constant

5. NUCLEAR FISSION AND FUSION

5.1 Nuclear Fission

A process in which a nuclear bombardment of the nucleus of a heavy nuclide by particles such as neutrons results in splitting of the nucleus into two smaller nuclei with the release of a large amount of energy

Example:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{148}_{57}La + ^{85}_{35}Br + 3^{1}_{0}n + \text{Energy Release}$$

A heavy nucleus (e.g., uranium-235) splits into two lighter nuclei with the release of a large amount of energy and free neutrons.

Used in **nuclear reactors** and **atomic bombs**.

Chain reaction can occur if released neutrons initiate further fissions.

5.2 Nuclear Fusion

Two light nuclei (e.g., hydrogen isotopes) combine to form a heavier nucleus.

Releases even **more energy** than fission.

Occurs naturally in **stars**, including the Sun.

Requires extremely high temperature and pressure.

6. COMPONENTS OF A NUCLEAR REACTOR

Component	Function
Nuclear Fuels	Supply the needed energy to power the nuclear reactor
Nuclear Moderator	Slow down fast-moving neutrons to cause further fission (e.g., graphite)
Control Rods	Absorb neutrons to prevent further fission (e.g., boron, cadmium)
Reactor Coolants	Carry heat produced by fission to external boiler and turbine; converted to electricity (e.g., liquid sodium, lead)
Turbine	Transfers heat from coolant to electricity
Cooling Towers	Dump excess heat that cannot be converted to energy due to the laws of thermodynamics

7. USES AND HAZARDS OF NUCLEAR ENERGY

Applications:

Field	Application
Medicine	Radiation therapy for cancer treatment; radioisotopes in diagnostic imaging (e.g., iodine-131 for thyroid imaging)
Industry	Non-destructive testing of materials using radioactive tracers; sterilization of medical equipment
Power Generation	Nuclear reactors generate electricity using controlled fission (e.g., Pressurized Water Reactor - PWR, Boiling Water Reactor - BWR)

Hazards

Exposure to high doses of radiation can cause cancer, genetic mutations, and acute radiation sickness

Radioactive waste disposal poses long-term environmental and health risk

Nuclear accidents (e.g., Chernobyl, Fukushima) can have catastrophic consequences

8. EINSTEIN'S MASS-ENERGY EQUIVALENCE AND BINDING ENERGY CALCULATION

The equation **$E = \Delta m \cdot c^2$** is used to calculate binding energy when the mass defect is known. Binding energy per nucleon is an important measure of nuclear stability.

Worked Example 1:

A radioactive isotope of Americium (Am-241) decays into a nucleus of Neptunium (Np-237) and an alpha (α) particle as shown in the nuclear equation below:

$$^{241}_{95}Am \rightarrow {}^{237}\_cNp + {}^{b}_{a}\alpha$$

Determine the values of a, b, and c.

Solution:

Recall that $^{b}_{a}\alpha = {}^{4}_{2}He$

$$^{241}_{95}Am \rightarrow {}^{237}\_cNp + {}^{4}_{2}He$$

(i) 95 = c + 2 → **c = 93**

(ii) Number of neutrons = 241 - 95 = **14**

Worked Example 2:

If a nucleus has a mass defect Δm = 0.0025 u, convert to kg and calculate the energy released.

Solution (in kg):

$$\Delta m = 0.0025 \times 1.66 \times 10^{-27}\ \text{kg} = 4.15 \times 10^{-30}\ \text{kg}$$

$$E = 4.15 \times 10^{-30} \times (3 \times 10^{8})^{2} = 3.735 \times 10^{-13}\ \text{J}$$

Alternative (using 1 u = 931 MeV):

$$E = 0.0025 \times 931 = 2.33\ \text{MeV}$$

Worked Example 3:

In a nuclear reaction, the mass defect is 2.0×10^{-6} g. Calculate the energy released, given that the velocity of light is 3.0×10^{8} m/s.

Solution:

$\Delta m = 2.0 \times 10^{-6}$ g $= 2.0 \times 10^{-9}$ kg

$c = 3 \times 10^{8}$ m/s

$$E = \Delta m \cdot c^2 = (2.0 \times 10^{-9}) \times (3 \times 10^{8})^2$$

$$= 2.0 \times 10^{-9} \times 9 \times 10^{16}$$

$$= 18 \times 10^{7}\ \text{J}$$

$$= \mathbf{1.8 \times 10^{8}\ J}$$

9. EVALUATION (WAEC/NECO PAST QUESTIONS)

(WAEC 2019 Q6): Explain the difference between mass number and atomic number of an element.

(NECO 2021 Q9): State two differences between nuclear fission and fusion.

(WAEC 2022 Q3): A helium nucleus has a mass defect of 0.0304 u. Calculate its binding energy. (1 u = 931 MeV

(WAEC 2018 Q10): Give one medical and one industrial application of nuclear energy.

(NECO 2020 Q4): What is an isotope? Give two examples of isotopes of hydrogen

10. ASSIGNMEN

Define the following term

Nuclide

Isotope

Binding energy

Differentiate between nuclear fission and nuclear fusion with one example each.

Calculate the binding energy in MeV of a lithium nucleus that has a mass defect of 0.035 u.